

Color and Texture-Changeable Modules for Pin-based Shape Display with Spring Structures

Mai Kagawa*
The University of Tokyo
Tokyo, Japan
kagawa@xlab.iii.u-tokyo.ac.jp

Tomomi Imamura*
The University of Tokyo
Tokyo, Japan
imamura@xlab.iii.u-tokyo.ac.jp

Yasuaki Kakehi
The University of Tokyo
Tokyo, Japan
kakehi@iii.u-tokyo.ac.jp

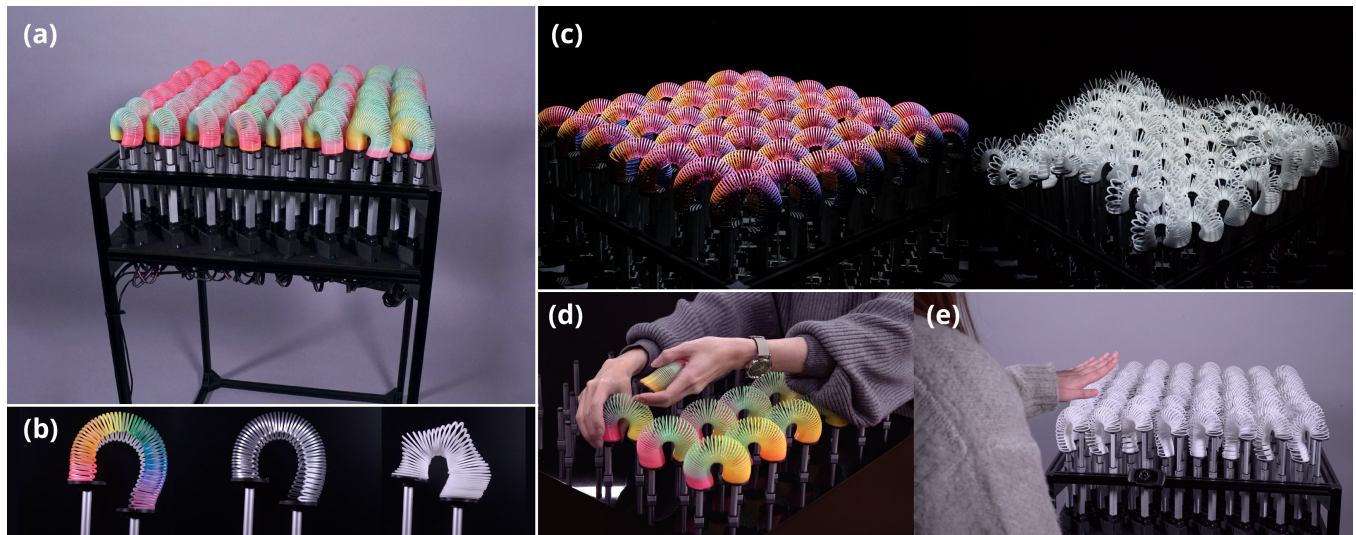


Figure 1: (a) Overall view of the implemented shape display with helical spring modules, (b) Example of a spring module, (c) Variations of texture representation by interchanging modules, (d) Module configuration that can be easily interchanged by attachments, (e) Examples of interactive applications.

Abstract

In this paper, we propose a modular system based on a spring structure that achieves texture variation expression using physical objects, with a configuration equivalent to a pin-based shape display. Using only the actuators as the driving source, the deformation of a single spring module attached in an arch shape by the linkage of two actuators achieves texture variation expression with simple wiring and control. With a reconfigurable structure using magnet attachments, users can easily change the expression by simply replacing the spring module. We propose a spring design method for creating texture changes through spring deformation, and demonstrate the system's expressive potential through design examples. We also introduce examples of interactive applications combined with a sensing system.

*Both authors contributed equally to this research.

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CCS Concepts

• Human-centered computing → Interaction devices.

Keywords

Pin-based Shape Display; Shape Changing Interfaces; Spring; Texture change

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1 Introduction and related work

Information representation using physical objects has gained significant attention in both Human-Computer Interaction (HCI) and artistic domains. In the field of HCI, concepts such as rendering screen-displayed information into tangible objects and facilitating interaction through deformable physical forms have been proposed to enhance sensory experiences and connect digital information through physical interaction [8, 9]. Based on these visions, shape-changing interfaces [21, 25] and shape displays [10] have been extensively studied. Among various shape-changing technologies, pin-based shape displays, which use arrays of linear actuators to

achieve 2.5-dimensional motion and shapes, are the most common method [3, 5, 11, 14, 16–20, 24, 26–29].

In parallel, the field of artistic expression has explored kinetic installations that integrate the material properties and textures of physical objects into interactive displays. Daniel Rozin’s “Wooden Mirror” and other mechanical mirrors [22, 23] utilize variations in material angles to render grayscale or binary images. These works incorporate the textures of materials such as wood, metal, and fabric into display systems, creating a visual effect that dynamically responds to human movement, like a mirror. These works highlight the potential for combining material textures and physical interactivity within an artistic context.

Examples of pin-based shape displays, such as inFORM [3], have demonstrated their potential to render dynamic physical forms and enable novel interaction possibilities. Other approaches exploit material properties, including inflatable actuators [4], magnetic actuators [29], and suspended objects actuated through vertical string displacement [2]. While alternative actuation methods have been explored, their expressive capabilities are often restricted to shape deformation. A common limitation among these systems is that dynamic shape changes and additional attributes, such as color and texture, are interdependent, which constrains the sensory richness of interaction. Techniques such as projection overlays [11] and embedded lighting [19, 20] enable independent control of visual imagery by modulating LED colors and physical shape transformations. While these approaches mitigate some interaction limitations, they still face challenges such as environmental dependency and wiring complexity.

In terms of texture representation, prior works have utilized fabrication techniques to create static tactile experiences. For instance, tactile displays simulating roughness or smoothness often rely on predefined material properties. Systems such as the Brixel Mirror[1] illustrate the potential of dynamic textures but are limited to binary states or predefined configurations. Current fabrication technologies, including 3D printing, enable multi-material designs, yet these implementations rarely achieve dynamic texture modulation. Integrating dynamic textures into shape-changing systems remains an open challenge, particularly in balancing between complexity and expressiveness.

This study addresses these limitations by introducing a helical spring module system designed for pin-based shape displays. Our helical spring module system leverages vertical actuator movements to enable dynamic texture and color transformations while maintaining a simple and modular design. The system allows for customizable and interchangeable modules, providing enhanced expressiveness and functionality.

In summary, we contributed:

- (1) A novel method to realize texture change expression on a shape display by controlling linear actuators by combining spring modules.
- (2) Policies of the spring design method for achieving texture change.
- (3) Simple texture customization with interchangeable modules.
- (4) Examples of interactive applications combined with sensing systems.

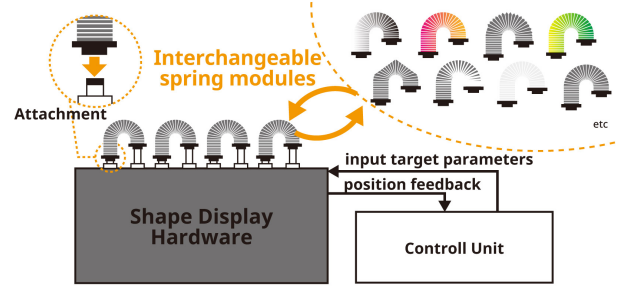


Figure 2: Overview of system configuration composed with interchangeable spring modules, shape display, control unit.

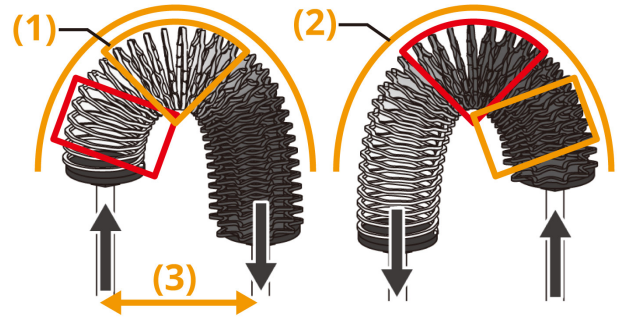


Figure 3: Basic principles for texture changes of helical spring pixels. Helical spring modules are suitable because (1) the visible portion of the spring transitions through the vertical movement of both ends, (2) the arch shape is maintained during the transition, and (3) the structural flexibility can absorb differences in distance between actuators.

2 Our Approach

In this study, we propose a method of combining one spring-structured driven module for two linear actuators to extend the representation of shape displays and to realize texture changes at the pixel level. As Fig. 2 shows the overall design configuration, our method can attach interchangeable modules to the linear actuators via attachments.

2.1 Basic Principles for Texture Changes

In this section, we describe the basic principles of representing texture change with spring modules. Many shape-changing displays represent 2.5-dimensional shapes using vertical actuator movement arranged in a grid. Our proposal extends these systems by enabling not only shape changes but also texture and color changes through linear actuator control. Each spring module, affixed to two actuators in an arch formation, functions as a single pixel. By mapping textures and colors within the spring, transitions are achieved via vertical movement, as shown in Fig. 3.

The scenario where the left or right actuator is elevated is shown in Fig. 3, highlighting the range perceived as a pixel in each state. As the vertex transitions, the visible range within the spring shifts, enabling texture changes by mapping different colors and textures within the module.

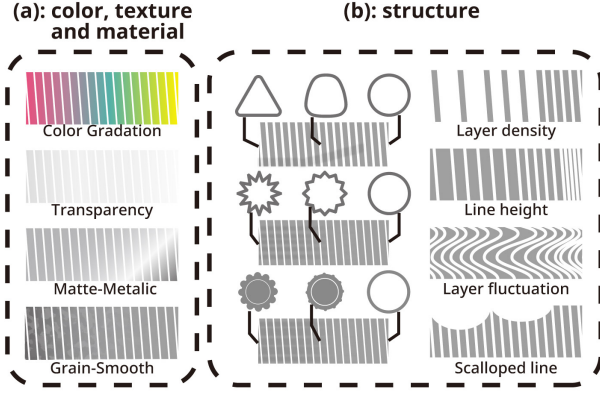


Figure 4: Two design policies for Adding Texture to Springs: (a) changing the color or the material, (b) changing the structure.



Figure 5: Examples of spring modules with texture designs.

There are three requirements for a module to apply vertical motion of linear actuators to texture changes: (1) it must deform through parallel motion, (2) maintain pixel shape during deformation, and (3) be robust to variations in scale and distance. Helical springs, widely studied for their deformability in HCI[6, 7], meet these conditions by allowing visible transitions with vertical movement, maintaining their arch shape, and absorbing spacing differences. In contrast, rigid modules require precise actuator spacing, increasing the risk of damage. Flexible modules adapt to various displays and simplify control, reducing breakage risks and enabling intuitive operation.

2.2 Design Guidelines for Adding Texture to Springs

In this section, we propose 2 policies as a design technique to give a spring a change in texture. The 2 policies are (a) changing the color, texture, and material of the surface, (b) changing the structure of the spring (Fig. 4). For instance, in (a), by painting different colors or using different materials in different layers, changes in color, transparency, luster, and smoothness can be created. In (b), the undulations appearing on the arch and arch shape can be varied by changing the cross-sectional shape of each layer of the spiral or

by changing the thickness of the layers, the distance between the gaps or the line width. These two policies are independent and can be reflected in the design at the same time, as shown in Fig. 5.

2.3 Interchangeability by Attachments

One of the advantages of a configuration that combines a group of actuators and passive modules [12, 13, 15, 16] is its ease of customization by simply interchanging the modules. Our approach adopts a modular system of linear actuators and interchangeable modules that are connected by attachments. Easily replaceable module configurations make it possible to individually customize only the modules to be installed. This allows the system to achieve a variety of texture representation with a variety of modules, even if it solely controls the vertical movement of a linear actuator.

3 Implementation

In this section, we describe the implementation of our concept. The implemented pin-based shape display, which is the basis for the shape change, the practiced design and production process for the spring modules, and the attachment parts to combine the two are described below.

3.1 Pin-based Shape Display

We implemented an array of 8×8 pixel modules shown in Fig. 6. The L12-100-50-6-I micro linear actuators from Actuonix were used as linear actuators to drive shape change and texture representation. These actuators operate on a 6V power supply, feature a compact overall length of 152 mm, offer a stroke length of 100 mm, and are capable of achieving a speed of 25 mm/s under no load conditions. These actuators were arranged in a configuration of 128 units. The distance between the paired actuator centers is 45 mm, the distance between the module centers is 50 mm, and the size of the top board is 600 mm $\times$ 600 mm. The actuators are fixed vertically by cases made with a 3D printer.

We implemented an electrical system to control the actuators, which serve as the driving source. The actuators use an RC Servo Interface Mode, enabling position control via pulse length. A 16-channel servo drive kit with NXP's PCA9685 servo controller and an Arduino Uno microcontroller was used. Eight servo controllers, managed by a master Arduino Uno via I2C communication, controlled up to 128 actuators. By pre-programming movements on the master microcontroller and switching target values over time, standalone operation was achieved.

3.2 Fabrication Method

This section explains how to create spring shapes and attach parts to them when making spring modules.

3.2.1 Spring Modules. The spring modules in this system were fabricated using three methods, as shown in Fig. 8. First, the attachments were adhered to commercially available toys for use as spring modules (Fig. 8 (a)). Second, a method was used where flat structures printed using an FDM 3D printer were immersed in hot water at approximately 70 degrees Celsius and then formed into spring shapes (Fig. 8 (b)). This approach is particularly suited for printing multicolored modules or creating modules with varying

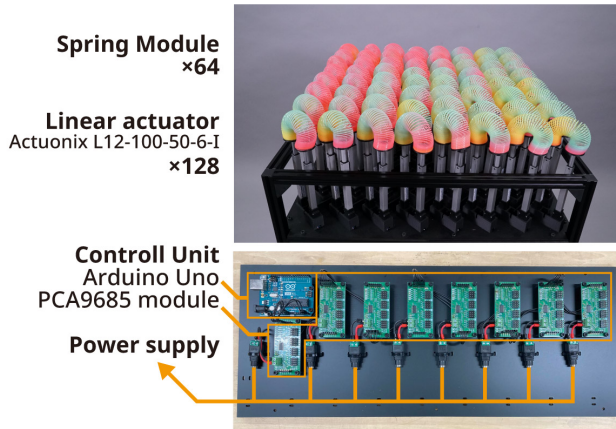


Figure 6: Side view of the device and control panel.

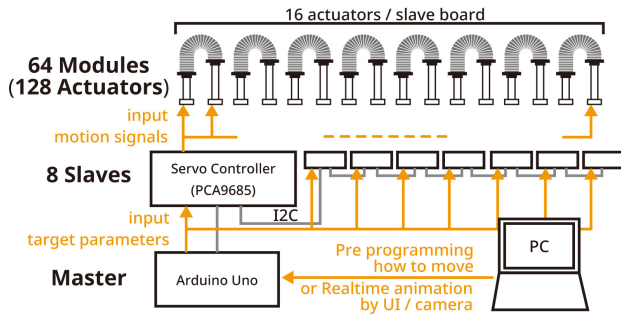


Figure 7: Overall configuration of the implemented shape display.

materials. Third, springs with partially modified structures were fabricated using an SLA 3D printer (Fig. 8 (c)).

3.2.2 Attachments. Attachments connecting actuators and modules must be easily detachable while securely holding the modules during operation to ensure synchronized movement with the actuators. To meet these requirements, attachment parts were designed with embedded neodymium magnets (cylindrical shape, 6 mm in diameter and 2 mm in height) in 3D-printed components, as shown in Fig. 9. Hexagonal pillars were used to prevent twisting during operation and were glued to both ends of the spring modules with a glue gun.

4 Application Example

4.1 Pre-programmed Texture and Shape Changes

We implemented a system in which the texture and overall shape of individual spring modules dynamically change based on pre-programmed movements. As shown in Fig. 10 (a), the system can simulate specific patterns by synchronizing actuator movements, enabling complex transitions between dense and sparse textures or altering the overall contour of the display.

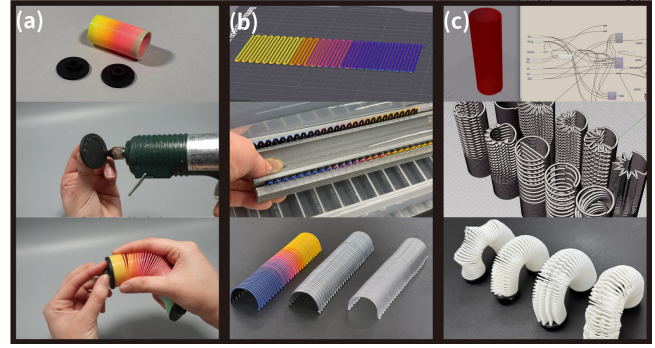


Figure 8: (a) Use of existing spring-shaped toys, (b) Color and material variation spring module using FDM-type 3D printing, (c) Structural change spring modules by photolithographic 3D printing.

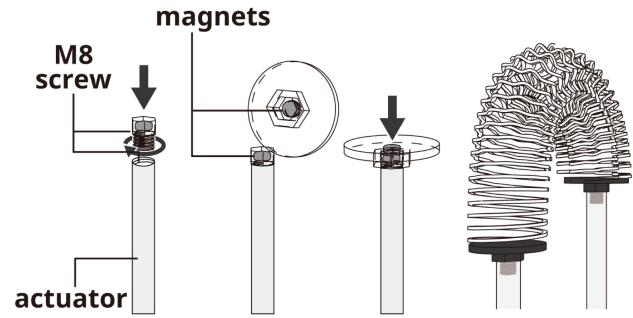


Figure 9: Attachments: Hexagonal shape and embedded magnets for secure retention and easy detachment.

4.2 Interactive Texture Mapping

A system was developed that tracks hand movements using a camera and changes the texture of the display based on the traced path (Fig. 10 (b)). The texture of the corresponding regions dynamically transitions, allowing localized changes based on the traced positions.

4.3 Dynamic Color Display

We implemented a display system that dynamically changes its overall color in response to the color of an object captured by a camera (Fig. 10 (c)). The detected color is processed in real time and the spring modules are adjusted to reflect the captured hue.

5 Conclusion

This paper proposes a system to achieve dynamic expression of texture variation with a configuration similar to traditional pin-based shape displays. By attaching an arch-shaped spring to two actuators and utilizing a linear actuator not only for overall shape changes but also for the state transitions of the module's texture, we proposed a novel approach to texture expression. We presented a simple actuator control system, multiple design methods for designing texture changes in the spring, and reconfigurable attachments



Figure 10: (a) Pre-programming movement, (b) Hand tracking interaction, (c) Dynamic Color Display.

that users can easily replace. Furthermore, we demonstrated the potential to expand the expressive power of the proposed method through interactive application examples that integrate sensing with shape display.

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References

- [1] BREAKFAST. 2018. Brixel Mirror. <https://breakfaststudio.com/works/brixel-mirror>. (Accessed on 1/24/2025).
- [2] Severin Engert, Konstantin Klamka, Andreas Peetz, and Raimund Dachselt. 2022. STRAIDE: A Research Platform for Shape-Changing Spatial Displays based on Actuated Strings. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (, New Orleans, LA, USA.) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 263, 16 pages. doi:10.1145/3491102.3517462
- [3] Sean Follmer, Daniel Leithinger, Alex Olwal, Akimitsu Hogue, and Hiroshi Ishii. 2013. inFORM: dynamic physical affordances and constraints through shape and object actuation. In *Proceedings of the 26th Annual ACM Symposium on User Interface Software and Technology* (St. Andrews, Scotland, United Kingdom) (UIST '13). Association for Computing Machinery, New York, NY, USA, 417–426. doi:10.1145/2501988.2502032
- [4] Juri Fujii, Takuya Matsunobu, and Yasuaki Kakehi. 2018. COLORISE: Shape- and Color-Changing Pixels with Inflatable Elastomers and Interactions. In *Proceedings of the Twelfth International Conference on Tangible, Embedded, and Embodied Interaction* (Stockholm, Sweden) (TEI '18). Association for Computing Machinery, New York, NY, USA, 199–204. doi:10.1145/3173225.3173228
- [5] John Hardy, Christian Weichel, Faisal Taher, John Vidler, and Jason Alexander. 2015. ShapeClip: Towards Rapid Prototyping with Shape-Changing Displays for Designers. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (Seoul, Republic of Korea) (CHI '15). Association for Computing Machinery, New York, NY, USA, 19–28. doi:10.1145/2702123.2702599
- [6] Liang He, Huaishu Peng, Michelle Lin, Ravikanth Konjeti, François Guimbretière, and Jon E. Froehlich. 2019. Ondulé: Designing and Controlling 3D Printable Springs. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (New Orleans, LA, USA) (UIST '19). Association for Computing Machinery, New York, NY, USA, 739–750. doi:10.1145/3332165.3347951
- [7] Liang He, Xia Su, Huaishu Peng, Jeffrey Ian Lipton, and Jon E. Froehlich. 2022. Kinergy: Creating 3D Printable Motion using Embedded Kinetic Energy. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) (UIST '22). Association for Computing Machinery, New York, NY, USA, Article 69, 15 pages. doi:10.1145/3526113.3545636
- [8] Hiroshi Ishii, Dávid Lakatos, Leonardo Bonanni, and Jean-Baptiste Labruno. 2012. Radical atoms: beyond tangible bits, toward transformable materials. *Interactions* 19, 1 (jan 2012), 38–51. doi:10.1145/2065327.2065337
- [9] Hiroshi Ishii and Brygg Ullmer. 1997. Tangible bits: towards seamless interfaces between people, bits and atoms. In *Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems* (Atlanta, Georgia, USA) (CHI '97). Association for Computing Machinery, New York, NY, USA, 234–241. doi:10.1145/258549.258715
- [10] Daniel Leithinger, Sean Follmer, Alex Olwal, and Hiroshi Ishii. 2015. Shape Displays: Spatial Interaction with Dynamic Physical Form. *IEEE Computer Graphics and Applications* 35, 5 (2015), 5–11. doi:10.1109/MCG.2015.111
- [11] Daniel Leithinger, David Lakatos, Anthony DeVincenzi, Matthew Blackshaw, and Hiroshi Ishii. 2011. Direct and gestural interaction with relief: a 2.5D shape display. In *Proceedings of the 24th Annual ACM Symposium on User Interface Software and Technology* (Santa Barbara, California, USA) (UIST '11). Association for Computing Machinery, New York, NY, USA, 541–548. doi:10.1145/2047196.2047268
- [12] Jiahao Li, Alexis Samoylov, Jeeun Kim, and Xiang 'Anthony' Chen. 2022. Roman: Making Everyday Objects Robotically Manipulable with 3D-Printable Add-on Mechanisms. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (, New Orleans, LA, USA.) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 272, 17 pages. doi:10.1145/3491102.3501818
- [13] Ken Nakagaki. 2020. Mechanical Shells: Physical Add-ons for Extending and Reconfiguring the Interactivities of Actuated TUIs. In *Adjunct Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology* (Virtual Event, USA) (UIST '20 Adjunct). Association for Computing Machinery, New York, NY, USA, 151–156. doi:10.1145/3379350.3415801
- [14] Ken Nakagaki, Daniel Fitzgerald, Zhiyao (John) Ma, Luke Vink, Daniel Levine, and Hiroshi Ishii. 2019. inFORCE: Bi-directional 'Force' Shape Display for Haptic Interaction. In *Proceedings of the Thirteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Tempe, Arizona, USA) (TEI '19). Association for Computing Machinery, New York, NY, USA, 615–623. doi:10.1145/3294109.3295621
- [15] Ken Nakagaki, Joanne Leong, Jordan L. Tappa, João Wilbert, and Hiroshi Ishii. 2020. HERMITS: Dynamically Reconfiguring the Interactivity of Self-propelled TUIs with Mechanical Shell Add-ons. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology* (Virtual Event, USA) (UIST '20). Association for Computing Machinery, New York, NY, USA, 882–896. doi:10.1145/3379337.3415831
- [16] Ken Nakagaki, Yingda (Roger) Liu, Chloe Nelson-Arzuaga, and Hiroshi Ishii. 2020. TRANS-DOCK: Expanding the Interactivity of Pin-based Shape Displays by Docking Mechanical Transducers. In *Proceedings of the Fourteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Sydney NSW, Australia) (TEI '20). Association for Computing Machinery, New York, NY, USA, 131–142. doi:10.1145/3374920.3374933
- [17] Ken Nakagaki, Udayan Umaphathi, Daniel Leithinger, and Hiroshi Ishii. 2017. AnimaStage: Hands-on Animated Craft on Pin-based Shape Displays. In *Proceedings of the 2017 Conference on Designing Interactive Systems* (Edinburgh, United Kingdom) (DIS '17). Association for Computing Machinery, New York, NY, USA, 1093–1097. doi:10.1145/3064663.3064670
- [18] Ken Nakagaki, Luke Vink, Jared Counts, Daniel Windham, Daniel Leithinger, Sean Follmer, and Hiroshi Ishii. 2016. Materialle: Rendering Dynamic Material Properties in Response to Direct Physical Touch with Shape Changing Interfaces. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 2764–2772. doi:10.1145/2858036.2858104
- [19] Ivan Poupyrev, Tatsushi Nashida, Shigeaki Maruyama, Jun Rekimoto, and Yasufumi Yamaji. 2004. Lumen: interactive visual and shape display for calm computing. In *ACM SIGGRAPH 2004 Emerging Technologies*. Association for Computing Machinery, New York, NY, USA, 17. doi:10.1145/1186155.1186173
- [20] Ivan Poupyrev, Tatsushi Nashida, and Makoto Okabe. 2007. Actuation and tangible user interfaces: the Vaucanson duck, robots, and shape displays. In *Proceedings of the 1st International Conference on Tangible and Embedded Interaction* (Baton Rouge, Louisiana) (TEI '07). Association for Computing Machinery, New York, NY, USA, 205–212. doi:10.1145/1226969.1227012
- [21] Majken K. Rasmussen, Esben W. Pedersen, Marianne G. Petersen, and Kasper Hornbæk. 2012. Shape-changing interfaces: a review of the design space and open research questions. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Austin, Texas, USA) (CHI '12). Association for Computing

- Machinery, New York, NY, USA, 735–744. doi:10.1145/2207676.2207781
- [22] Daniel Rozin. 1999. Wooden Mirror. <https://www.smoothware.com/danny/woodenmirror.html>. (Accessed on 1/24/2025).
- [23] Daniel Rozin. 2023. RGB Peg Mirror. In *ACM SIGGRAPH 2023 Art Gallery*. Association for Computing Machinery, New York, NY, USA. doi:10.1145/3588428.3593821
- [24] Alexa F. Siu, Eric J. Gonzalez, Shenli Yuan, Jason Ginsberg, Allen Zhao, and Sean Follmer. 2017. shapeShift: A Mobile Tabletop Shape Display for Tangible and Haptic Interaction. In *Adjunct Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology* (Québec City, QC, Canada) (UIST '17 Adjunct). Association for Computing Machinery, New York, NY, USA, 77–79. doi:10.1145/3131785.3131792
- [25] Miriam Sturdee and Jason Alexander. 2018. Analysis and Classification of Shape-Changing Interfaces for Design and Application-based Research. *ACM Comput. Surv.* 51, 1, Article 2 (Jan. 2018), 32 pages. doi:10.1145/3143559
- [26] Ryo Suzuki, Junichi Yamaoka, Daniel Leithinger, Tom Yeh, Mark D. Gross, Yoshihiro Kawahara, and Yasuaki Kakehi. 2018. Dynablock: Dynamic 3D Printing for Instant and Reconstructable Shape Formation. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology* (, Berlin, Germany,) (UIST '18). Association for Computing Machinery, New York, NY, USA, 99–111. doi:10.1145/3242587.3242659
- [27] Ryo Suzuki, Clement Zheng, Yasuaki Kakehi, Tom Yeh, Ellen Yi-Luen Do, Mark D. Gross, and Daniel Leithinger. 2019. ShapeBots: Shape-changing Swarm Robots. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (New Orleans, LA, USA) (UIST '19). Association for Computing Machinery, New York, NY, USA, 493–505. doi:10.1145/3332165.3347911
- [28] John Tiab, Sebastian Boring, Paul Strohmeier, Anders Markussen, Jason Alexander, and Kasper Hornbæk. 2018. Tiltstacks: composing shape-changing interfaces using tilting and stacking of modules. In *Proceedings of the 2018 International Conference on Advanced Visual Interfaces* (Castiglione della Pescaia, Grosseto, Italy) (AVI '18). Association for Computing Machinery, New York, NY, USA, Article 44, 5 pages. doi:10.1145/3206505.3206530
- [29] Kentaro Yasu. 2022. MagneShape: A Non-electrical Pin-Based Shape-Changing Display. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) (UIST '22). Association for Computing Machinery, New York, NY, USA, Article 71, 12 pages. doi:10.1145/3526113.3545645